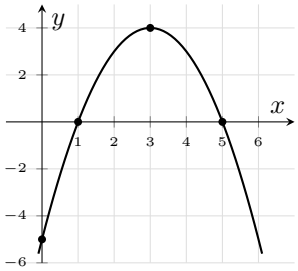
Lesson 1: Zeros, Roots and x-Intercepts

Assignment

1. The graph of $y = -x^2 + 6x - 5$ is shown. The marked points have integer coordinates. State:



(a) whether it opens up or down, and the vertex

(b) the axis of symmetry and the maximum or minimum value

(c) the x-intercepts and the y-intercept

(d) the zeros of the function and the roots of $-x^2 + 6x - 5 = 0$

2. Find the roots of each equation.

(a) $5x(x + 4) = 0$ (b) $2x^2 - 14x + 24 = 0$ (c) $9x^2 - 16 = 0$

(d) $3x^2 + 5x = 2$ (e) $x^3 - 3x^2 = 10x$ (f) $4x^2 + 1 = 4x$

3. Find the zeros of each function.

(a) $f(x) = 2x - 9$ (b) $g(x) = 49x^2 - 36$ (c) $h(x) = -2(3x-4)(x+5)$ (d) $P(x) = x(x+6)(4x-1)$

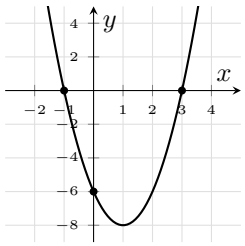
4. For each function, determine (i) the zeros and (ii) the y-intercept of its graph.

(a) $f(x) = 3x^2 - 12x$ (b) $f(x) = -x^2 + x + 20$

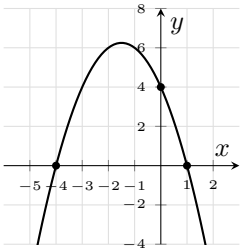
(c) $f(x) = 2x^2 + 3x - 9$ (d) $f(x) = 6x^2 - 13x - 5$

5. The graph of each quadratic function is shown. The marked points have integer coordinates. State the zeros and the y-intercept, then write the function in factored form.

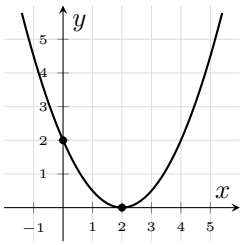
(a)



(b)



(c)



6. State the number of x-intercepts of each parabola, and explain.

(a) It opens up; its vertex is $(-2, 5)$.

(b) It opens down; its vertex is $(4, 1)$.

(c) Its vertex is $(3, 0)$.

(d) $y = -x^2 - 9$

7. Use a graphing calculator.

(a) Find the exact zeros of $f(x) = 10x^2 - 9x - 9$ and write $f(x)$ in factored form.

(b) Find the exact zeros of $g(x) = 2x^3 - x^2 - 8x + 4$ and write $g(x)$ in factored form.

(c) Find the zeros of $y = 2x^2 + 5x - 4$ and of $y = -x^2 + 4x + 7$, to the nearest hundredth.

8. Write each quadratic function in factored form, then in general form.

(a) zeros 2 and 6; passes through $(4, -8)$

(b) zeros -5 and 1; y-intercept 10

(c) zeros $-\frac{1}{2}$ and 3; passes through $(1, -6)$

(d) a double zero at -4; passes through $(-2, -6)$

9. Compare the graphs of $y = (x + 1)(x - 4)$ and $y = -3(x + 1)(x - 4)$. What do they have in common, and how are they different?

10. A diver jumps from a platform. Her height above the water after t seconds is $h(t) = -5t^2 + 5t + 10$ metres.

(a) Find the zeros of h. Which one has meaning here, and what does it tell you?

(b) What does the y-intercept of the graph of h represent?

(c) When is she 10 m above the water again?

11. A club sells hoodies. Its profit is $P(x) = -2x^2 + 140x - 2000$ dollars when each hoodie costs x dollars.

(a) Find the zeros of P (the *break-even* prices).

(b) Between which prices does the club make a profit? Explain using the graph.

(c) At what price is the profit greatest? (Use the axis of symmetry.)

12. The function $f(x) = x^2 + bx + c$ has zeros -3 and 7. Determine b and c.

13. A quadratic function has zeros -2 and 6, and its maximum value is 8. Write the function in factored form.

14. The zeros of $f(x) = -3(x+2)(5x-4)$ are

A. 2 and $-\frac{4}{5}$ **B.** -2 and $\frac{4}{5}$ **C.** -2 and $\frac{5}{4}$ **D.** -3, -2 and $\frac{4}{5}$

15. The roots of $2x(x-3) = 8$ are

A. 0 and 3 **B.** 4 and 11 **C.** -4 and 1 **D.** 4 and -1

16. Which function has exactly one zero?

A. $y = 4x^2 - 9$ **B.** $y = 4x^2 + 9$ **C.** $y = 4x^2 - 12x + 9$ **D.** $y = 4x^2 - 12x$

17. Numerical Response The zeros of $f(x) = 6x^2 - x - 15$ are m and n, where $m > n$. The value of $m - n$, to the nearest hundredth, is _____.

18. Numerical Response The graph of $y = a(x-2)(x-8)$ passes through $(4, -24)$. The y-intercept of the graph is _____.

Lesson 2: Graphing $y = a(x - p)^2 + q$

Assignment

1. Describe how the graph of each function is related to the graph of $y = x^2$.

(a) $y = (x - 7)^2$ (b) $y = x^2 + 9$ (c) $y + 6 = x^2$

(d) $y = (x + 4)^2 - 1$ (e) $y - 3 = (x - 5)^2 + 2$ (f) $y = (x + m)^2 + n, \ \ m, n > 0$

2. Write the equation of the image of $y = x^2$ after each translation.

(a) 9 units left (b) 4 units down (c) 2 units right and 6 units up (d) h units left and k units down

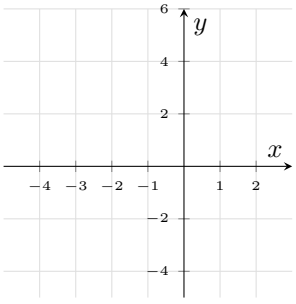
3. The point $(-3, 9)$ is on the graph of $y = x^2$. Find its image after each translation.

(a) 5 units right (b) 4 units left and 10 units down

4. Complete the table.

Function	Vertex	Max/min value	Axis	Domain	Range
(a) $y = x^2 - 8$					
(b) $y = (x + 5)^2$					
(c) $y + 1 = (x - 3)^2 + 4$					
(d) $y = (x - h)^2 + k$					

5. Sketch $y = (x + 1)^2 - 4$ by hand on the grid. State the vertex, the minimum value, the domain and the range.



6. After a horizontal and a vertical translation, the image of $y = x^2$ has its vertex at $(-5, 3)$. Describe the translations and write the equation of the image.

7. Describe how the graph of each function is related to the graph of $y = x^2$.

(a) $y = 5x^2$

(b) $y = -\frac{1}{4}x^2$

(c) $y = -2(x-6)^2 + 1$

(d) $y = \frac{2}{3}(x+3)^2 - 4$

(e) $3y = (x-1)^2 - 9$

(f) $-y = (x+2)^2$

8. The transformations are applied to $y = x^2$ in the order given. Write the equation of each image.

(a) a vertical stretch by a factor of 3 about the x-axis and a reflection in the x-axis

(b) a vertical stretch by a factor of $\frac{1}{2}$ about the x-axis and a translation of 7 units up

(c) a reflection in the x-axis, a vertical stretch by a factor of 5, and a translation of 2 units right and 4 units down

(d) a vertical stretch by a factor of $\frac{3}{4}$ and a translation of 6 units left and 1 unit up

(e) a vertical stretch by a factor of c ($c > 0$), a reflection in the x-axis, and a translation of d units left and e units up

9. The point $(2, 4)$ is on $y = x^2$. Find its image after each set of transformations.

(a) a reflection in the x-axis, then a translation of 6 units up

(b) a vertical stretch by a factor of $\frac{1}{4}$ about the x-axis

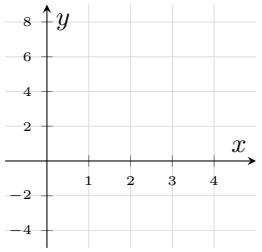
(c) a vertical stretch by a factor of 3, then a translation of 1 unit left and 5 units down

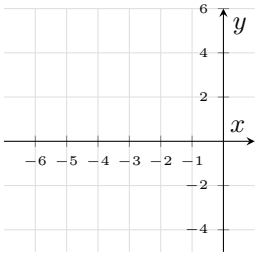
10. These transformations are applied to $y = x^2$ in order: a reflection in the x-axis, a vertical stretch by a factor of 2 about the x-axis, and a translation of 4 units right and 3 units up.

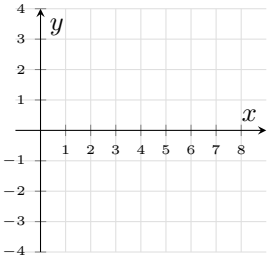
(a) Write the equation of the image after each transformation.

(b) The point $(6, k)$ is on the final graph. Determine k.

11. Sketch each graph using its vertex and the step pattern.
(a) $y = 3(x-2)^2 - 4$ (b) $y = -(x+3)^2 + 5$ (c) $y = \frac{1}{4}(x-4)^2 - 3$







12. Sam graphs $y = 3(x-1)^2 + 2$ by translating $y = x^2$ 1 unit right and 2 units up, and then stretching vertically by a factor of 3 about the x-axis. What vertex does he get? Explain his error and give the correct vertex.

13. Write the equation of the image of $y = x^2$ after a vertical stretch by the factor a given (a negative a includes a reflection) and a translation that moves the vertex to the point given.
(a) $a = 2$, vertex $(-1, 5)$ (b) $a = -\frac{1}{3}$, vertex $(6, 0)$ (c) $a = -5$, vertex $(0, -2)$

14. The graph of $y = x^2$ is translated so that its vertex is at $(-n, m)$, where $m > 0$ and $n > 0$. The equation of the image is
A. $y = (x-n)^2 + m$ **B.** $y = (x+n)^2 + m$ **C.** $y = (x+m)^2 + n$ **D.** $y = (x-m)^2 - n$

15. The point $(-1, 1)$ is on $y = x^2$. Its image on the graph of $y = -2(x-3)^2 + 5$ is
A. $(2, 3)$ **B.** $(-4, 3)$ **C.** $(2, 7)$ **D.** $(2, -12)$

16. Which graph is narrower than the graph of $y = x^2$ and opens down?
A. $y = -0.4x^2 + 3$ **B.** $y = 2.5(x+1)^2$ **C.** $y = -3(x-2)^2$ **D.** $y = -(x+5)^2 - 6$

17. Numerical Response The graph of $y = x^2$ is stretched vertically by a factor of $\frac{1}{2}$ about the x-axis, reflected in the x-axis, and translated 2 units right and 9 units up. The point $(5, k)$ is on the final graph. The value of k, to the nearest tenth, is _____.

18. Numerical Response The graph of $y = a(x-p)^2 + q$ is the image of $y = x^2$ after a vertical stretch by a factor of 3 about the x-axis, a reflection in the x-axis, and a translation of 2 units right and 7 units up. The value of $a + p + q$ is _____.

Lesson 3: Characteristics of a Quadratic Function in Vertex Form

Assignment

1. Complete the table. (The domain of each function is $\{x \mid x \in \mathbb{R}\}$.)

Function	Vertex	Axis of symmetry	Max or min value	Range
(a) $y = 5(x-7)^2 - 2$				
(b) $y = -(x+8)^2 + 3$				
(c) $y = -\frac{1}{4}x^2 + 9$				
(d) $y = 1.5(x+0.5)^2$				
(e) $y = -6(x-1)^2 - 4$				

2. Write each equation in the form $y = a(x-p)^2 + q$. Then state the vertex and the maximum or minimum value.
(a) $y + 6 = 4(x-3)^2$ (b) $2y = -(x+5)^2 + 10$ (c) $y = -1 + 3(x+7)^2$ (d) $y - 4 = -(x-9)^2$

3. Determine the y-intercept of each graph.
(a) $y = 3(x+2)^2 - 7$ (b) $y = -(x-6)^2 + 20$ (c) $y = \frac{1}{4}(x-8)^2 + 1$ (d) $y = -2(x+0.5)^2 + 3$

4. Without graphing, state the number of x-intercepts of each graph. Give a reason.
(a) $y = -2(x-3)^2 + 8$ (b) $y = 7(x+1)^2 + 0.5$ (c) $y = -(x-4)^2$

(d) $y = 0.3x^2 - 12$ (e) $y = -5(x+2)^2 - 1$

5. Determine the x-intercepts algebraically.

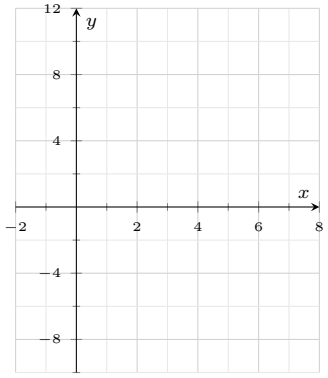
(a) $y = (x + 5)^2 - 49$ (b) $y = 3(x - 2)^2 - 75$ (c) $y = -2(x + 1)^2 + 32$

(d) $y = \frac{1}{4}(x - 6)^2 - 1$ (e) $y = 5(x - 3)^2 + 20$ (f) $y = -(x - 9)^2$

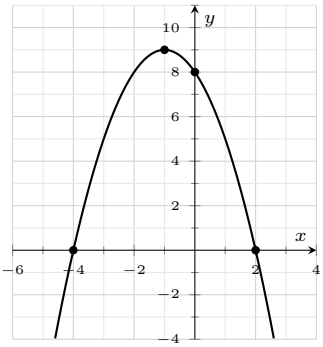
6. Determine the x-intercepts as exact values in simplest form, then to the nearest hundredth.
(a) $y = (x - 1)^2 - 7$ (b) $y = -2(x + 3)^2 + 16$ (c) $y = 3(x - 4)^2 - 10$

7. Analyze $y = 2(x - 3)^2 - 8$, then sketch its graph.

Vertex	
Axis of symmetry	
Max or min value	
Domain	
Range	
y-intercept	
x-intercepts	



8. The graph of a quadratic function is shown. The marked points have integer coordinates.



(a) State the vertex and the equation of the axis of symmetry.

(b) State the maximum or minimum value, the domain and the range.

(c) State the x- and y-intercepts.

(d) The equation is $y = -(x+1)^2 + 9$. Verify the x-intercepts algebraically.

9. Use symmetry.

(a) The graph of $y = a(x-4)^2 + q$ has an x-intercept at 10. Determine the other x-intercept.

(b) A parabola has vertex $(-2, 5)$ and passes through $(1, -13)$. Name another point on it.

(c) The graph of $y = ax^2 + q$ has an x-intercept at -3.5. Determine the other x-intercept.

10. Write an equation in the form $y = a(x-p)^2 + q$ for a parabola with the given features. (Many answers are possible.)

(a) vertex $(-3, 5)$, opens down

(b) axis $x = 2$, minimum value -6

(c) vertex $(1, -4)$ and no x-intercepts

(d) axis $x = -7$ and range $\{y \mid y \leq 0, y \in \mathbb{R}\}$

11. Kai says that $y = 3(x+2)^2 - 5$ has vertex $(2, -5)$ and no x-intercepts because q is negative. Identify both errors and give the correct vertex and number of x-intercepts.

12. A fountain nozzle sprays water in a parabolic arc. The height of the water above the surface of the pool is $h = -0.5(x-3)^2 + 5$ metres, where x is the horizontal distance in metres from the nozzle.

(a) What is the maximum height of the water, and at what horizontal distance does it occur?

(b) How high above the pool surface is the nozzle?

(c) How far from the nozzle does the water hit the pool surface? Give the answer to the nearest hundredth of a metre

and explain why one root is rejected.

(d) State the domain and range in this context.

13. A soccer ball is kicked from the ground. Its height after t seconds is $h = -5(t - 1.2)^2 + 7.2$ metres.

(a) What is the maximum height, and when does it occur?

(b) Show that the ball starts on the ground.

(c) How long is the ball in the air?

(d) At what times is the ball 5.4 m high?

14. The vertex of the graph of $y + 2 = -(x - 6)^2$ is

A. $(-6, 2)$ **B.** $(6, 2)$ **C.** $(6, -2)$ **D.** $(-6, -2)$

15. The range of $y = -3(x + 2)^2 + 5$ is

A. $\{y \mid y \leq 5, y \in \mathbb{R}\}$

B. $\{y \mid y \geq 5, y \in \mathbb{R}\}$

C. $\{y \mid y \leq -2, y \in \mathbb{R}\}$

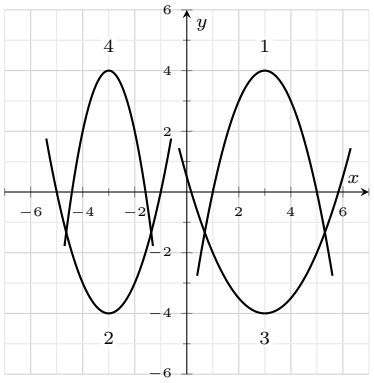
D. $\{y \mid y \geq -3, y \in \mathbb{R}\}$

16. Which function has no x-intercepts?

A. $y = 2(x - 1)^2 - 3$ **B.** $y = -(x + 4)^2 + 1$ **C.** $y = -3(x - 2)^2 - 5$ **D.** $y = \frac{1}{2}(x + 3)^2$

17. Numerical Response Four parabolas are numbered on the grid. Match the equations with the graphs.

Graph number of $y = \frac{1}{2}(x-3)^2 - 4$: first digit
Graph number of $y = -2(x+3)^2 + 4$: second digit
Graph number of $y = -(x-3)^2 + 4$: third digit
Graph number of $y = (x+3)^2 - 4$: fourth digit
(Record the four digits from left to right.)



18. Numerical Response The y-intercept of the graph of $y = -2(x+3)^2 + 50$ is _____.

Lesson 4: General Form and Factored Form: Intercepts, Axis and Vertex

Assignment

1. Write each function in general form.

(a) $y = (x + 5)(x - 3)$

(b) $y = -2(x - 4)(x - 1)$

(c) $y = \frac{1}{3}(x + 6)(x - 3)$

(d) $y = -(x + 2)^2 + 7$

(e) $y = 4(x - 0.5)^2 - 1$

(f) $y = (3x - 1)(x + 2)$

2. For each function, state the zeros, the axis of symmetry, the vertex and the y-intercept.

(a) $y = (x - 4)(x + 2)$

(b) $y = -(x + 9)(x + 1)$

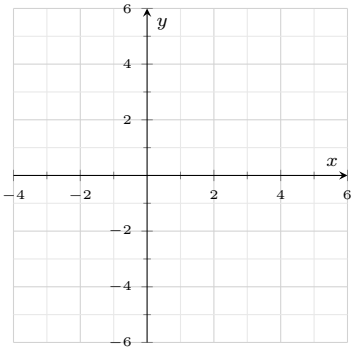
(c) $y = 4x(x + 3)$

(d) $y = -2(x - 3)(x - 8)$

(e) $y = (2x + 1)(x - 4)$

3. Analyze $y = \frac{1}{2}(x + 2)(x - 4)$, then sketch its graph.

Zeros	
Axis of symmetry	
Vertex	
y-intercept	
Range	

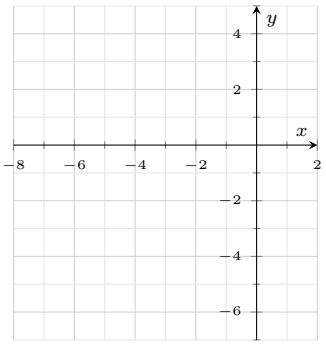


4. Complete the table.

Function	Axis	Vertex	Max or min	y-int	Range
(a) $y = x^2 + 6x - 2$					
(b) $y = -2x^2 + 8x + 3$					
(c) $y = 3x^2 - 18x + 20$					
(d) $y = \frac{1}{2}x^2 - 3x + 2$					
(e) $y = -x^2 + 7x$					

5. Factor each function. Then state the zeros, the axis of symmetry and the vertex.
(a) $y = x^2 + 4x - 21$ (b) $y = -2x^2 + 2x + 12$ (c) $y = 3x^2 - 12$

6. Consider $y = -x^2 - 6x - 5$.
(a) State the y-intercept, the axis of symmetry and the vertex.



(b) Write the function in factored form and state the zeros.

(c) Sketch the graph and state its range.

7. For each function, find the vertex, write the function in vertex form, then determine the x-intercepts exactly and to the nearest hundredth.
(a) $y = x^2 - 6x + 4$ (b) $y = -3x^2 + 6x + 2$

8. Show that $y = -(x - 2)^2 + 9$, $y = -x^2 + 4x + 5$ and $y = -(x - 5)(x + 1)$ are the same function. Then state the vertex, the y-intercept, the zeros and the range, and say which form you read each one from.

9. Determine the unknown coefficient.
(a) The graph of $y = x^2 + bx + 3$ has axis of symmetry $x = -4$. Determine b and the vertex.

(b) The function $y = -2x^2 + 12x + c$ has a maximum value of 11. Determine c.

(c) The graph of $y = ax^2 - 6x + 1$ has axis of symmetry $x = -1$. Determine a, the vertex, and whether it is a maximum or a minimum.

10. Use symmetry.

(a) The points $(-1, 8)$ and $(5, 8)$ are on a parabola. State its axis of symmetry.

(b) A parabola has zeros -7 and 3 and passes through $(1, 12)$. Name another point on the parabola.

11. A ball is thrown from the roof of a shed. Its height after t seconds is $h = -5t^2 + 15t + 20$ metres.

(a) Write h in factored form.

(b) How high is the roof, and when does the ball land?

(c) Determine the maximum height and when it occurs.

(d) State the domain and range in this context.

12. A stone arch is modelled by $h = -0.1x(x - 30)$, where h is the height in metres and x is the horizontal distance in metres from one foot of the arch.

(a) How wide is the arch at the ground?

(b) How high is the top of the arch?

(c) How high is the arch 5 m from a foot?

13. Jordan says the axis of symmetry of $y = 2x^2 + 12x - 5$ is $x = 3$ and the vertex is $(3, 49)$. Explain the error and find the correct vertex.

14. The axis of symmetry of the graph of $y = 3x^2 - 12x + 7$ is

A. $x = 2$ **B.** $x = -2$ **C.** $x = 4$ **D.** $x = -4$

15. The vertex of the graph of $y = -(x + 3)(x - 5)$ is

A. $(1, 16)$ **B.** $(1, -16)$ **C.** $(-1, 12)$ **D.** $(0, 15)$

16. The range of $y = 2x^2 + 8x + 3$ is

A. $\{y \mid y \geq 3\}$ **B.** $\{y \mid y \leq -5\}$ **C.** $\{y \mid y \geq -5\}$ **D.** $\{y \mid y \geq -2\}$

17. Numerical Response The maximum value of $y = -2x^2 + 16x - 5$ is _____.

18. Numerical Response The vertex of the graph of $y = 4(x - 1)(x - 6)$ is (p, q). The value of $p - q$, to the nearest tenth, is _____.

Lesson 5: Writing a Quadratic Equation from Its Vertex and a Point

Assignment

1. A parabola has vertex $(2, -1)$ and passes through the point $(4, 11)$.

(a) Write its equation in vertex form.

(b) Write its equation in general form.

(c) Determine the x- and y-intercepts of the graph. Give the x-intercepts to the nearest hundredth, and check them with a graphing calculator.

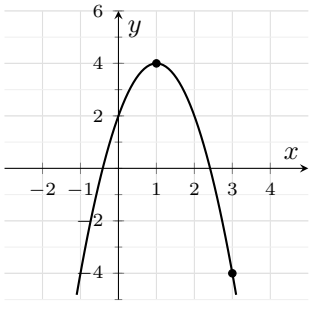
2. Write an equation in vertex form for the parabola with the given vertex that passes through the given point.

(a) vertex $(-5, 2)$, point $(-3, 10)$ (b) vertex $(4, 7)$, point $(1, -11)$ (c) vertex $(0, -3)$, point $(6, 9)$

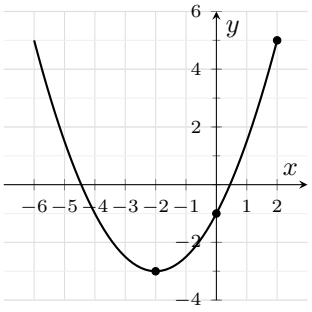
(d) vertex $(-2, 0)$, point $(2, -4)$ (e) vertex $(6, -1)$, y-intercept 17 (f) vertex $(-3, 5)$, point $(-7, -3)$

3. Write an equation of the form $y = a(x - p)^2 + q$ for each graph.

(a)



(b)



4. The graph of a quadratic function has its vertex at $(-1, 8)$, and one of its x-intercepts is 1.
(a) Determine the equation in vertex form and in general form.

(b) State the other x-intercept, the equation of the axis of symmetry, the domain and the range.

5. A parabola has vertex $(3, -6)$ and passes through $(1, -4)$. Determine its equation, then its x-intercepts as exact values and to the nearest hundredth, and its y-intercept.

6. The point $(4, -3)$ is on the graph of $y = ax^2 + q$. Name another point on the graph without finding a or q, and explain how you know.

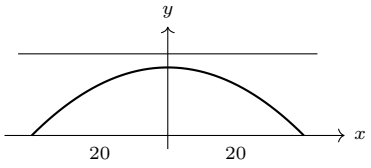
7. A parabola has axis of symmetry $x = -4$ and passes through $(-4, 7)$ and $(-2, -1)$. Write its equation in vertex form.

8. A parabola has axis of symmetry $x = 1$ and passes through $(-1, 5)$ and $(4, 20)$. Write its equation in vertex form and state its range.

9. Write each equation in factored form, and determine the vertex.
(a) zeros -3 and 1, through $(-1, -8)$ (b) zeros 0 and 8, through $(2, -6)$ (c) zeros -5 and 3, y-intercept 6

10. The graph of a quadratic function touches the x-axis only at $x = 2$, and its y-intercept is -12. Determine its equation in vertex form and in general form.

11. A road bridge is supported by a parabolic arch that spans 40 m at its base and rises to 10 m at its centre.



(a) Place the origin on the base, directly below the top of the arch. Determine the equation of the arch.

(b) How high is the arch 8 m in from one end of the base?

(c) How wide is the arch at a height of 7.5 m?

12. A golf ball is hit from the ground. Its path is a parabola that reaches a greatest height of 30 m at a horizontal distance of 60 m from the tee.

(a) Using the tee as the origin, determine the equation of the path.

(b) How high is the ball when it is 100 m from the tee horizontally? Answer to the nearest tenth.

(c) How far from the tee does it land?

13. The jet of water from a fountain leaves a nozzle 1.2 m above the ground. It reaches its greatest height, 3.2 m, at a horizontal distance of 2 m from the nozzle. Using the ground below the nozzle as the origin, determine the equation of the jet, and how far from the nozzle (horizontally) the water hits the ground, to the nearest hundredth of a metre.

14. A quadratic function has a minimum value of -5 at $x = -2$, and its graph passes through $(0, 3)$ and $(1, k)$. The value of k is

A. -3 **B.** -0.5 **C.** 13 **D.** 31

15. A parabola has x-intercepts -2 and 4 and passes through $(2, 4)$. Its equation is

A. $y = -\frac{1}{2}(x - 2)(x + 4)$

B. $y = -2(x + 2)(x - 4)$

C. $y = \frac{1}{2}(x + 2)(x - 4)$

D. $y = -\frac{1}{2}(x + 2)(x - 4)$

16. Numerical Response A parabola has vertex $(4, -9)$ and passes through $(10, 0)$. When its equation is written as $y = a(x - p)^2 + q$, the value of a, to the nearest hundredth, is _____.

17. Numerical Response The points $(-2.3, 5)$ and $(k, 5)$, $k \neq -2.3$, are on the graph of $y = a(x - 1.5)^2 + q$. The value of k, to the nearest tenth, is _____.

Lesson 6: Completing the Square

Assignment

1. Determine the number that must be added to make each expression a perfect square, then write the result as a square.
(a) $x^2 + 14x$ (b) $x^2 - 22x$ (c) $x^2 + 9x$

(d) $x^2 - x$ (e) $x^2 + 0.8x$ (f) $x^2 - \frac{4}{5}x$

2. Write each function in vertex form by completing the square.
(a) $y = x^2 + 6x + 1$ (b) $y = x^2 - 12x + 40$ (c) $y = x^2 + 2x - 15$

(d) $y = x^2 - 7x + 10$ (e) $y = x^2 + x + 1$ (f) $y = x^2 + 11x - 2$

3. Write $f(x) = x^2 - 16x + 50$ in vertex form. Hence state the vertex, the axis of symmetry, the minimum value and the range.

4. Write each function in vertex form.
(a) $y = 2x^2 + 8x + 3$ (b) $y = 3x^2 - 30x + 70$ (c) $y = 5x^2 + 10x$

(d) $y = 2x^2 - 6x + 1$ (e) $y = 4x^2 + 4x - 3$

5. Write each function in vertex form.
(a) $y = -x^2 - 6x + 2$ (b) $y = -2x^2 + 16x - 25$ (c) $y = 7 + 12x - 3x^2$

(d) $y = \frac{1}{2}x^2 + 4x + 5$ (e) $y = -\frac{1}{4}x^2 + 3x - 4$ (f) $y = 10x - 5x^2$

6. Complete the square. State the vertex, and whether the function has a maximum or a minimum value, and what it is.
(a) $f(x) = 2x^2 + 20x + 53$ (b) $g(x) = -3x^2 + 18x - 20$

7. Complete the square to determine the zeros of each function as exact values and to the nearest hundredth.
(a) $f(x) = x^2 - 6x - 11$ (b) $f(x) = 2x^2 + 4x - 7$

8. Complete the square to show that the graph of $y = x^2 + 6x + 13$ has no x-intercepts.

9. A model rocket's height, in metres, t seconds after launch is $h = -5t^2 + 30t + 2$. Complete the square to find its maximum height and when it occurs.

10. A rectangular pen is fenced on three sides, with a barn wall as the fourth side. If each side perpendicular to the wall is w metres, the area is $A = -2w^2 + 36w$. Complete the square to find the greatest possible area and the value of w that gives it.

11. A student wrote $y = 3x^2 + 12x - 1$ in vertex form. Identify the step with the error, explain it, and write the correct vertex form.

Step 1: $y = 3(x^2 + 4x) - 1$ Step 3: $y = 3(x + 2)^2 - 4 - 1$

Step 2: $y = 3(x^2 + 4x + 4 - 4) - 1$ Step 4: $y = 3(x + 2)^2 - 5$

12. Write $y = 3x^2 - 12x + k$ in vertex form, in terms of k. For what value of k is the vertex on the x-axis?

13. Complete the square for $y = ax^2 + bx + c$ (in terms of a, b and c), and show that the axis of symmetry is $x = -\dfrac{b}{2a}$.

14. The graph of $y = x^2 + bx + c$ has vertex $(-4, 3)$. Determine b and c.

15. The range of $y = 2x^2 - 6x + 1$ is

A. $\{y \mid y \geq -1.25, y \in \mathbb{R}\}$ **B.** $\{y \mid y \geq 5.5, y \in \mathbb{R}\}$ **C.** $\{y \mid y \geq -3.5, y \in \mathbb{R}\}$ **D.** $\{y \mid y \leq -3.5, y \in \mathbb{R}\}$

16. The maximum value of $f(x) = 3x(3 - x)$ is

A. $\frac{3}{2}$ **B.** $\frac{27}{4}$ **C.** $\frac{9}{4}$ **D.** $\frac{27}{2}$

17. Numerical Response A ball's height, in metres, t seconds after it is tossed is $h = -2t^2 + 7t + 1$. Its maximum height, to the nearest hundredth of a metre, is _____.

18. Numerical Response When $y = 3x^2 + 12x + 17$ is written in the form $y = a(x - p)^2 + q$, the value of $a + p + q$ is _____.

Lesson 7: The Quadratic Formula

Assignment

1. Solve $x^2 + 2x - 15 = 0$ (a) by factoring, (b) with the quadratic formula.

2. Solve $6x^2 - 7x - 3 = 0$ (a) by decomposition, (b) with the quadratic formula.

3. Solve by completing the square. Then check with the quadratic formula.
(a) $x^2 - 6x + 2 = 0$ (b) $2x^2 + 8x + 3 = 0$

4. Solve with the quadratic formula. Give exact roots in simplest form.
(a) $x^2 - 8x + 4 = 0$ (b) $x^2 + 10x + 18 = 0$ (c) $4x^2 - 4x - 7 = 0$

(d) $3x^2 + 4x - 2 = 0$ (e) $x^2 + 3x + 5 = 0$ (f) $5x^2 - 2x - 1 = 0$

5. Solve with the quadratic formula. Give the roots to the nearest tenth.
(a) $3x^2 + x - 5 = 0$ (b) $5x^2 - 8x + 2 = 0$ (c) $0.5x^2 - 1.2x - 3 = 0$

6. Determine the zeros of each function, exactly and to the nearest hundredth.
(a) $y = x^2 - 14x + 9$ (b) $y = -2x^2 + 6x + 5$

(c) State the x-intercepts of the graph in (b), to the nearest hundredth, and describe how to check them with a graphing calculator.

7. A student solved $3x^2 + 6x - 2 = 0$ as shown. Identify the two errors and give the correct exact roots.

$$x = \frac{-6 \pm \sqrt{6^2 - 4(3)(-2)}}{2(3)}$$
$$= \frac{-6 \pm \sqrt{36 - 24}}{6} = \frac{-6 \pm \sqrt{12}}{6}$$
$$= \frac{-6 \pm 2\sqrt{3}}{6} = -1 \pm 2\sqrt{3}$$

8. Choose an efficient method, name it, and solve. Give exact roots.
(a) $7x^2 - 28 = 0$ (b) $x^2 + 9x + 14 = 0$ (c) $2(x + 1)^2 - 18 = 0$

(d) $x^2 - 5x + 1 = 0$ (e) $9x^2 = 6x$ (f) $2x^2 + x - 4 = 0$

9. Write each equation in standard form, then solve. Give exact roots in simplest form.
(a) $x(x-4) = 3x - 5$ (b) $(x+3)^2 = 2x + 10$ (c) $(3x-1)(x+2) = (x+1)(x+4)$

(d) $\frac{x^2}{4} + \frac{x}{3} - 1 = 0$ (e) $x - \frac{5}{x} = 2, \ x \neq 0$ (f) $20x^2 + 40x = 15$

10. A soccer ball is kicked from a height of 1.5 m. Its height after t seconds is $h = -4.9t^2 + 12t + 1.5$ metres.
(a) When does the ball land, to the nearest tenth of a second?

(b) At what times is the ball 6 m above the ground, to the nearest tenth of a second?

11. A rectangular dog run is enclosed by 28 m of fencing and has an area of 42 m^2. Determine its exact dimensions, then give them to the nearest tenth of a metre.

12. The legs of a right triangle differ by 3 cm and the hypotenuse is 12 cm. Determine the exact length of the shorter leg, then both legs to the nearest hundredth.

13. The arithmetic series $2 + 5 + 8 + \ldots$ has $S_n = \frac{n(3n+1)}{2}$.
(a) Use the quadratic formula to find how many terms give a sum of 100.

(b) Can the sum of the series ever be exactly 150? Explain.

14. To solve $3x^2 - 5x = 2$, the correct substitution into the quadratic formula is

A. $\dfrac{5 \pm \sqrt{25 - 24}}{6}$ **B.** $\dfrac{-5 \pm \sqrt{25 + 24}}{6}$ **C.** $\dfrac{5 \pm \sqrt{25 + 24}}{6}$ **D.** $\dfrac{5 \pm \sqrt{-25 + 24}}{6}$

15. The zeros of $y = 4x^2 - 8x + 1$ are

A. $1 \pm 4\sqrt{3}$ **B.** $\dfrac{2 \pm \sqrt{3}}{2}$ **C.** $\dfrac{-2 \pm \sqrt{3}}{2}$ **D.** $2 \pm \sqrt{3}$

16. Numerical Response The larger root of $2x^2 - 7x + 1 = 0$, to the nearest hundredth, is _____.

17. Numerical Response The roots of $x^2 + 8x + 3 = 0$ can be written as $m \pm \sqrt{n}$, where m and n are integers. The value of $m + n$ is _____.

Lesson 8: The Discriminant

Assignment

1. Determine the value of the discriminant.
(a) $x^2 + 3x + 8 = 0$ (b) $4x^2 - 12x + 9 = 0$ (c) $2x^2 - 5x - 3 = 0$ (d) $5 - 3x^2 = x$

2. Determine the nature of the roots without solving.
(a) $x^2 - 7x + 10 = 0$ (b) $9x^2 + 6x + 1 = 0$ (c) $3x^2 - 2x + 5 = 0$

(d) $x^2 + 4x - 6 = 0$ (e) $2x(x - 3) = -5$ (f) $16x^2 = 24x - 9$

3. How many x-intercepts does the graph of each function have?
(a) $y = x^2 - 10x + 25$ (b) $y = -2x^2 + 3x + 7$ (c) $y = 3x^2 - x + 2$ (d) $y = -4(x - 2)^2 + 3$

4. Use the discriminant to decide whether the roots are rational or irrational. If rational, solve by factoring.
(a) $12x^2 - x - 6 = 0$ (b) $3x^2 + 4x - 5 = 0$ (c) $8x^2 - 10x + 3 = 0$

5. Consider $x^2 - 10x + k = 0$.
(a) For what values of k does it have (i) two distinct real roots, (ii) two equal real roots, (iii) no real roots?

(b) Give an integer value of k for which the roots are rational and not zero. Show that it works.

6. For what values of k does $-2x^2 + 4x + k = 0$ have (a) two distinct real roots, (b) two equal real roots, (c) no real roots?

7. Each equation has two equal real roots. Determine every possible value of k, and the root for each value.
(a) $x^2 + kx + 25 = 0$ (b) $4x^2 - kx + 1 = 0$ (c) $kx^2 - 12x + 4 = 0$

8. For what values of k does each equation have real roots?
(a) $kx^2 + 4x + 2 = 0$ (b) $3x^2 - 12x + 2k = 0$

9. For what values of m does $(m - 2)x^2 + 6x + 3 = 0$ have no real roots?

10. Show that each equation has two distinct real roots for every real number k.
(a) $2x^2 - kx - 1 = 0$ (b) $x^2 + 2kx + k^2 - 9 = 0$ (also show the roots are rational when k is an integer)

11. Explain why $ax^2 + bx + c = 0$ always has two distinct real roots when a and c have opposite signs.

12. A soccer ball is kicked so that its height after t seconds is $h = -4.9t^2 + 14t + 0.5$ metres. Use the discriminant to decide whether it reaches (a) 12 m, (b) 10 m.

13. A theatre's ticket revenue, in dollars, is $R = -20x^2 + 400x + 3000$, where x is the number of \$1 price increases. Use the discriminant to decide whether the revenue can be (a) \$5000, (b) \$5200. Explain what (a) tells you about the

maximum revenue.

14. The vertex of the parabola $y = 2x^2 - 8x + c$ lies on the x-axis. Determine c and the vertex.

15. Which equation has two distinct real roots?

A. $x^2 + 2x + 5 = 0$ **B.** $4x^2 - 4x + 1 = 0$ **C.** $3x^2 - x - 2 = 0$ **D.** $x^2 + 9 = 0$

16. The equation $x^2 + kx + 4 = 0$ has no real roots when

A. $k < 4$ **B.** $-4 < k < 4$ **C.** $k < -4$ or $k > 4$ **D.** $-2 < k < 2$

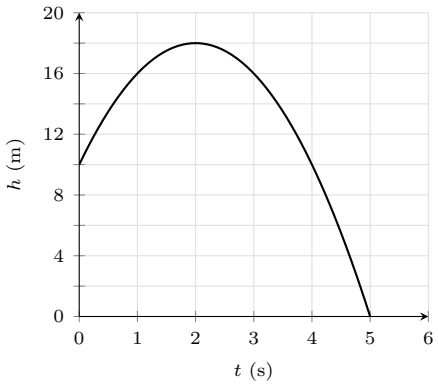
17. Numerical Response The equation $3x^2 - 12x + k = 0$ has two equal real roots when k is _____.

18. Numerical Response The greatest integer value of k for which $2x^2 + 7x + k = 0$ has two distinct real roots is _____.

Lesson 9: Maximum and Minimum Problems: Graphical Approach

Assignment

1. A stone is thrown from the top of a cliff. Its height above the beach in metres after t seconds is $h(t) = -2t^2 + 8t + 10$, graphed at the right. Use the graph.



(a) From what height was the stone thrown?

(b) What is its maximum height, and when is it reached?

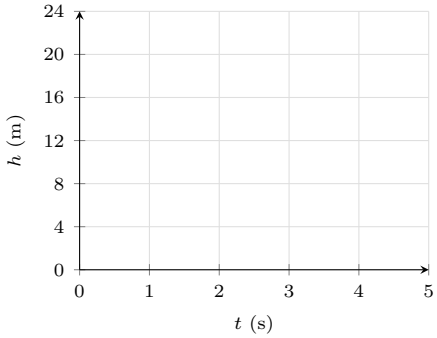
(c) When does it land on the beach?

(d) At what other time is it 10 m high?

(e) State the domain and range in this context.

2. A football is punted. Its height in metres above the field after t seconds is $h(t) = -4.9t^2 + 19.6t + 0.8$, $t \geq 0$. Use a graphing calculator. Answer to the nearest hundredth where necessary.

(a) Sketch the part of the graph that fits the context.



(b) How high was the ball when it was kicked?

(c) How high was it 1 s after the kick?

(d) What is the maximum height, and which coordinate of the vertex is it?

(e) When is the maximum height reached, and which coordinate of the vertex is it?

(f) The ball is not caught. When does it hit the field?

(g) The domain was given as $t \geq 0$. Write a more accurate domain, and the range.

(h) When is the ball 15 m above the field?

3. A skateboard bowl has a parabolic cross-section. Its height in metres relative to the ground (negative means below the ground), x metres from the left rim, is $y = 0.08x^2 - 0.96x$. Answer to the nearest hundredth of a metre.

(a) How deep is the bowl 3 m from the left rim?

(b) What is the greatest depth, and how far from the left rim is it?

(c) How wide is the bowl at ground level?

(d) How far from the left rim is the bowl exactly 2 m deep?

4. A climbing gym has 480 members who each pay \$60 a month. The owner learns that for every \$5 increase in the monthly fee, 20 members would leave. The fee changes in steps of \$5.

(a) Let x be the number of \$5 increases. Write expressions for the fee and the number of members, and the revenue function $R(x)$.

(b) Use a graphing calculator to find the fee that gives the maximum monthly revenue.

(c) How many members would there be, and what is the maximum revenue?

(d) How much more is this than the current monthly revenue?

5. The manager of the gym in question 4 suggests *lowering* the fee instead: for every \$3 decrease, 40 new members would join.

(a) Let x be the number of \$3 decreases. Write the revenue function and find the fee, the number of members and the revenue at the maximum.

(b) Which plan would you advise the owner to follow? Explain.

6. A fish farm could sell its stock today: 5000 kg of fish at \$6.00 per kg. For each week it waits, the fish gain a total of 400 kg, but the market price falls by \$0.20 per kg. The farm sells at the end of a whole week.

(a) Write the revenue function $R(w)$, where w is the number of weeks the farm waits.

(b) After how many whole weeks should the farm sell to earn the most? What is that revenue?

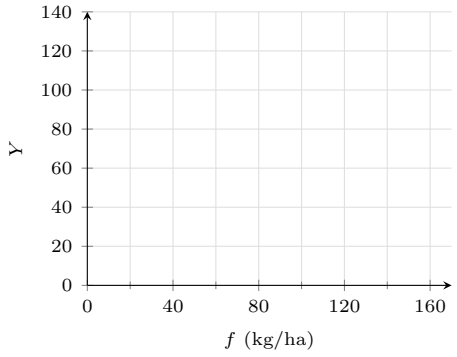
7. A farm tests different amounts of fertilizer on test plots and records the crop yield.

Fertilizer, f (kg/ha)	0	40	80	120	160
Yield, Y (bushels/acre)	31	100	127	104	28

The data are modelled by $Y(f) = -0.015f^2 + 2.4f + 30$.

(a) Plot the data on the grid and join the points with a smooth curve.

(b) Use the model to find the amount of fertilizer that gives the greatest yield, and that yield.



(c) Which gives the greater yield by the model: 20 kg/ha or 150 kg/ha?

8. A rectangular parking lot is fenced on three sides; the fourth side is a building. With 400 m of fencing and x metres for each side perpendicular to the building, the area in square metres is $A(x) = x(400 - 2x)$.

(a) Use a graphing calculator to find the maximum area and the dimensions that give it.

(b) For which values of x is the area 15 000 m^2?

9. A bicycle maker's average cost per bike, in dollars, when it makes n bikes a week is $C(n) = 0.002n^2 - 1.2n + 250$, for $50 \leq n \leq 550$.

(a) How many bikes should it make each week to keep the average cost as low as possible? What is the lowest average cost?

(b) For what numbers of bikes is the average cost \$100 or less?

10. Sam says, "A ball is in the air for exactly twice the time it takes to reach its maximum height."

(a) Test Sam's claim on $h(t) = -5t^2 + 20t$ and on $h(t) = -5t^2 + 20t + 25$.

(b) Explain, using the symmetry of the parabola, when Sam's claim is true.

11. A ball's height is given by $h(t) = -5t^2 + 15t + 2$. The feature of its graph that tells when the ball hits the ground is

A. the y-intercept

B. the x-coordinate of the vertex

C. the positive t-intercept

D. the y-coordinate of the vertex

12. The height of a drone flying a curved path is modelled by $h(t) = -0.5t^2 + 6t + 4$. To find when the drone is 20 m high with a graphing calculator, a student should

A. find the zeros of $Y_1 = -0.5X^2 + 6X + 4$

B. find the maximum of Y_1

C. graph $Y_2 = 20$ and find the intersections with Y_1

D. evaluate Y_1 at $X = 20$

Use the following information for questions 13 and 14.

A streaming service models its number of subscribers, S (millions), x years after the start of 2015 by $S(x) = -0.4x^2 + 6.4x + 13.5$.

13. Numerical Response According to the model, the number of subscribers peaks in the year _____.

14. Numerical Response The greatest number of subscribers, in millions to the nearest tenth, is _____.

Lesson 10: Maximum and Minimum Problems: Algebraic Approach

Assignment

1. Complete the square. State the vertex and the maximum or minimum value.

(a) $y = x^2 - 10x + 31$

(b) $y = -3x^2 + 18x - 4$

(c) $y = 2x^2 + 6x + 1$

(d) $y = -\frac{1}{2}x^2 + 4x - 3$

2. Determine the vertex using $x = -\dfrac{b}{2a}$, the zeros or the vertex formula. State the maximum or minimum value.

(a) $y = -0.5x^2 + 6x + 3$

(b) $y = -2(x - 4)(x + 10)$

(c) $f(x) = 4x^2 - 3x + 2$

(d) $y = 3x^2 + 7x$

3. A soccer ball is kicked. Its height in metres after t seconds is $h(t) = -5t^2 + 18t + 0.4$.

(a) Complete the square to write $h(t)$ in vertex form.

(b) How high is the ball after 1 s?

(c) State the maximum height and when it is reached.

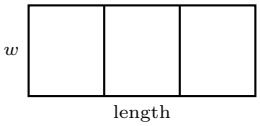
(d) How high, in centimetres, was the ball when it was kicked?

(e) When does the ball land? Answer to the nearest tenth of a second.

4. (a) Two numbers have a sum of 16. Determine the numbers for which the sum of their squares is a minimum, and that minimum.

(b) The sum of a number and twice a second number is 20. Determine the numbers whose product is a maximum, and that product.

5. A rectangular field is divided into three pens by two fences parallel to its width, as shown. A total of 160 m of fencing is used. Let w be the width. Determine the dimensions that give the greatest total area, and that area.



6. A rectangular storage yard is built against a warehouse wall, so only three sides need fencing. The side parallel to the wall uses fencing that costs \$12 per metre; the two ends cost \$8 per metre. The budget is \$960.

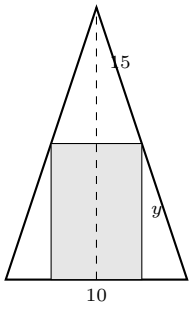
(a) Let each end be x metres. Show that the side parallel to the wall is $80 - \frac{4}{3}x$ metres.

(b) Determine the dimensions that give the largest yard, and its area.

7. A rectangle is drawn inside an isosceles triangle with base 10 cm and height 15 cm. The bottom of the rectangle lies on the base and its top corners touch the other two sides.

(a) Let the height of the rectangle be y. Use similar triangles to show that its width is $10 - \frac{2}{3}y$.

(b) Determine the dimensions of the rectangle with the greatest area, and that area.



8. A school sells 600 yearbooks at \$40 each. For every \$2 increase in price, 20 fewer yearbooks would sell. Let x be the number of \$2 increases.

(a) Write the revenue function.

(b) Determine the price that gives the maximum revenue, the number sold and the revenue.

9. A ski hill sells 2000 day passes a day at \$90 each. Research shows that each \$3 decrease in price brings 100 more skiers. Determine the price that gives the maximum daily revenue, the number of skiers and the revenue.

10. A wire 80 cm long is cut into two pieces. Each piece is bent into a square. Where should the wire be cut so that the total area of the two squares is as small as possible? What is that area?

11. A stone is thrown straight up. Its height in metres after t seconds is $h(t) = 24t - 5t^2$. When is the stone 20 m above the ground? Answer to the nearest tenth of a second, and explain why there are two answers.

12. The perimeter of a rectangular lot is 58 m and its area is 204 m^2.

(a) Let the length be x. Write a quadratic equation and solve it to find the dimensions.

(b) What is the greatest area a rectangle with this perimeter can have?

13. A path along two edges of a rectangular park totals 25 m. A diagonal shortcut across the park is 19 m. Determine the lengths of the two edges, to the nearest tenth of a metre.

14. The greatest possible area of a rectangle with a perimeter of 36 cm is

A. 324 cm^2 **B.** 81 cm^2 **C.** 80 cm^2 **D.** 72 cm^2

15. A ball's height in metres is $h(t) = -5t^2 + 40t + 3$. The maximum height is reached after

A. 4 s **B.** 8 s **C.** -4 s **D.** 83 s

16. For the revenue function $R(x) = (20 + x)(300 - 10x)$, where x is the number of \$1 increases on a \$20 price, the price that gives the maximum revenue is

A. \$5 **B.** \$25 **C.** \$30 **D.** \$50

17. Numerical Response A rectangular garden is fenced on three sides with 120 m of fencing; the fourth side is a house wall. The maximum area of the garden, in square metres, is _____.

18. Numerical Response A beanbag is tossed upward. Its height in metres after t seconds is $h(t) = -5t^2 + 8t + 1.5$. Its maximum height, in metres to the nearest tenth, is _____.

Lesson 11: Unit 5 Review

Assignment

1. Consider the function $y = 2x^2 - 2x - 12$.

(a) Determine the zeros of the function and the x-intercepts of its graph.

(b) How are your answers to (a) related to the roots of $2x^2 - 2x - 12 = 0$?

(c) Write the equation, in factored form, of the quadratic function with zeros 3 and -4 whose graph passes through $(2, 12)$.

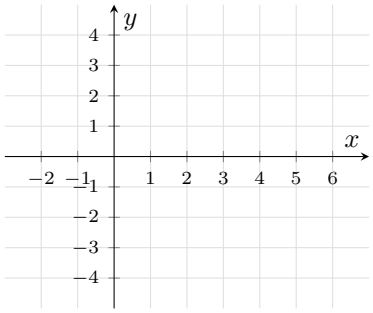
2. Describe how the graph of $y = x^2$ is transformed to give each graph.

(a) $y = (x - 4)^2 + 1$ (b) $y = -\frac{1}{2}(x + 2)^2$ (c) $y = 3(x + 1)^2 - 5$

(d) The graph of $y = x^2$ is stretched vertically by a factor of 2, reflected in the x-axis, and translated 3 units right and 4 units up. Write the equation of the new graph.

(e) The point $(3, 9)$ is on the graph of $y = x^2$. To which point is it mapped on the graph of $y = 3(x + 1)^2 - 5$?

3. Sketch the graph of $y = \frac{1}{2}(x - 2)^2 - 4$ using its vertex and the step pattern. Then state its range and the number of x-intercepts.



4. Complete the table for each function.

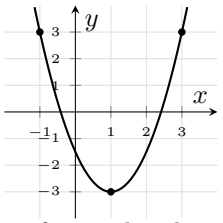
	$y = 3(x-2)^2 - 12$	$y = -\frac{1}{2}(x+4)^2 - 3$
Vertex		
Axis of symmetry		
Direction of opening		
Maximum or minimum value		
Domain		
Range		
y-intercept		
x-intercepts		

5. For each function, state the y-intercept, the zeros, the equation of the axis of symmetry, the vertex and the range.
(a) $y = -x^2 + 6x + 7$

(b) $y = 2(x+5)(x-1)$ (also write it in general form)

6. Write the equation of each quadratic function.
(a) Vertex $(-3, 2)$, passing through $(-1, 10)$

(b) The graph shown (vertex form)



(c) Axis of symmetry $x = 2$, range $\{y \mid y \geq -8, y \in \mathbb{R}\}$, y-intercept -6. Then find its x-intercepts.

(d) Zeros $\frac{1}{2}$ and 4, y-intercept 8. Give factored and general form.

7. The arch under a footbridge is a parabola. It spans 30 m across a stream, and its highest point is 6 m above the water. Place the origin at water level directly below the highest point.

(a) Write the equation of the arch.

(b) A tour boat 10 m wide and 4 m tall passes through the centre of the arch. Will it fit? Show why.

(c) How wide is the arch at a height of 4 m above the water, to the nearest tenth of a metre?

8. Complete the square to write each function in vertex form. State the vertex and the maximum or minimum value.

(a) $y = x^2 + 10x + 18$

(b) $y = 3x^2 - 18x + 22$

(c) $y = -2x^2 - 8x + 1$

(d) $y = \frac{1}{2}x^2 + 3x - 1$

9. Mila wrote $y = 2x^2 + 12x + 5$ in vertex form like this:

$$y = 2(x^2 + 6x + 9 - 9) + 5 = 2(x + 3)^2 - 9 + 5 = 2(x + 3)^2 - 4$$

Explain her error, give the correct vertex form, and show a quick check that proves her answer is wrong.

10. Solve using the quadratic formula. Give exact answers in simplest form for (a), (b) and (d), and round (c) to the nearest hundredth.

(a) $x^2 + 4x - 1 = 0$

(b) $2x^2 - 2x - 1 = 0$

(c) $5x^2 = 3x + 1$

(d) $x(x + 3) = 2(x + 5)$

11. Solve each equation exactly. Choose an efficient method and name it.

(a) $2(x - 3)^2 - 14 = 0$

(b) $6x^2 - x - 2 = 0$

(c) $x^2 + 2x + 5 = 0$

12. (a) Without solving, determine the nature of the roots: (i) $9x^2 - 6x + 1 = 0$ (ii) $3x^2 - 5x - 4 = 0$ (iii) $2x^2 + 3x + 4 = 0$

(b) Determine the values of k for which $2x^2 + kx + 8 = 0$ has equal roots.

(c) Determine the values of k for which $x^2 - 4x + k = 0$ has no real roots.

(d) Determine the values of m for which $mx^2 + 4x - 2 = 0$ has two distinct real roots.

(e) How many x-intercepts does the graph of $y = -3x^2 + 2x - 1$ have? Explain using the discriminant and the shape of the graph.

13. A bakery's daily profit, P dollars, from selling x dozen cookies is modelled by $P = -0.4x^2 + 36x - 450$. Use a graphing calculator (or algebra) to answer.

(a) How many dozen should be sold to make the maximum profit? What is the maximum profit?

(b) How many dozen must be sold to break even ($P = 0$)?

(c) For what numbers of dozens is the profit at least \$200?

(d) What does the P-intercept mean in this context?

14. A farmer has 120 m of fencing to build a rectangular pen along the side of a barn. The barn wall forms one side and needs no fence. Determine the dimensions that give the maximum area, and the maximum area.

15. The length of a rectangle is 4 cm more than its width, and its diagonal is 12 cm. Determine the width, exactly and to the nearest tenth of a centimetre.

16. Which statement is **false**?

A. The roots of $x^2 - 2x - 8 = 0$ are the zeros of the function $y = x^2 - 2x - 8$.

B. If $x = 6$ is a root of $ax^2 + bx + c = 0$, then $x - 6$ is a factor of $ax^2 + bx + c$.

C. If $x + 5$ is a factor of $ax^2 + bx + c$, then the graph of $y = ax^2 + bx + c$ has an x-intercept at $(5, 0)$.

D. A quadratic function whose vertex is on the x-axis has exactly one zero.

17. The graph of $y = x^2$ is translated 3 units right and 2 units down. The equation of the new graph is

A. $y = (x + 3)^2 - 2$ **B.** $y = (x - 3)^2 - 2$ **C.** $y = (x - 2)^2 + 3$ **D.** $y = (x - 3)^2 + 2$

18. Numerical Response The graph of $y = a(x - 3)^2 - 7$ passes through the point $(5, 5)$. The value of a is _____.

19. The range of $y = -3(x+1)^2 + 5$ is

A. $\{y \mid y \geq 5, y \in \mathbb{R}\}$ **B.** $\{y \mid y \leq 5, y \in \mathbb{R}\}$ **C.** $\{y \mid y \leq -1, y \in \mathbb{R}\}$ **D.** $\{y \mid y \geq -1, y \in \mathbb{R}\}$

20. The vertex of the graph of $y = x^2 + 8x + 10$ is

A. $(4, -6)$ **B.** $(-4, -6)$ **C.** $(-4, 10)$ **D.** $(-8, 10)$

21. Numerical Response The function $y = x^2 - 10x + 32$ can be written in the form $y = (x-a)^2 + b$. Write the value of a in the first box and the value of b in the second box.

22. A parabola passes through the points $(-1, 7)$ and $(5, 7)$. The equation of its axis of symmetry is

A. $x = 2$ **B.** $x = 3$ **C.** $x = 7$ **D.** $x = 6$

23. Which function has a graph with **no** x-intercepts?

A. $y = 2(x-3)^2 - 1$ **B.** $y = -2(x-3)^2 + 1$ **C.** $y = 2(x-3)^2 + 1$ **D.** $y = -2(x+3)^2$

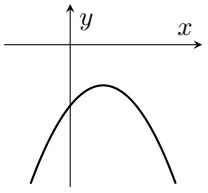
24. The roots of $2x^2 - 6x + 3 = 0$ are

A. $\dfrac{3 \pm \sqrt{3}}{2}$ **B.** $\dfrac{-3 \pm \sqrt{3}}{2}$ **C.** $\dfrac{3 \pm \sqrt{15}}{2}$ **D.** $\dfrac{6 \pm \sqrt{3}}{4}$

25. Numerical Response The positive root of $3x^2 - 4x - 9 = 0$, to the nearest tenth, is _____.

26. The graph of $y = ax^2 + bx + c$ is shown. Which statement is true?

A. $a > 0$ and $b^2 - 4ac > 0$
B. $a > 0$ and $b^2 - 4ac < 0$
C. $a < 0$ and $b^2 - 4ac > 0$
D. $a < 0$ and $b^2 - 4ac < 0$



27. Two numbers have a difference of 10. The minimum possible value of their product is

A. -25 **B.** 0 **C.** -24 **D.** 25

28. Numerical Response A rectangle has a perimeter of 52 cm. The maximum possible area of the rectangle, in square centimetres, is _____.
